

Sensational Stories: The Role of Narrative Characteristics in Distinguishing Real and Fake News and Predicting their Spread

Anne Hamby¹, Hongmin Kim², Francesca Spezzano²

¹Department of Marketing, Boise State University, 1910 University Dr., Boise, ID 83725

²Department of Computer Science, Boise State University, 1910 University Dr., Boise, ID 83725

Sensational Stories: The Role of Narrativity in Distinguishing Real and Fake Media Content and Predicting their Spread

Abstract

Media content – whether real or fake – is often presented in a story format. While research has examined some of the linguistic differences between real and fake news articles, less attention has been given to the ways in which they vary in terms of their narrativity. The current work leverages text analysis to build on recent work in the consumer field examining narrativity in predicting messages' persuasiveness. We conduct three studies in the context of real and fake news to examine how they differ in terms of their narrative content and discourse features. We further examine how variation in narrativity predicts the likelihood that the article is shared on social media, an outcome of importance to marketers and persuasion agents.

Keywords: Fake news, narrative, sharing behavior, social media

Sensational Stories: The Role of Narrativity in Distinguishing Real and Fake News and Predicting their Spread

1. Introduction

The proliferation of online social media has facilitated consumers' ability to share information and stay informed about trending events. While lowered barriers to communication can democratize access to information, they also enable the dissemination of content that is dubious and, in some cases, intended to mislead. Such content is often called fake news. Fake news is cited as one of the most pressing concerns for modern democracies, inciting calls from governments and grant agencies to understand its origins, distributions, and effects (Born & Edgington, 2017; National Science Foundation, 2022; Rockefeller Foundation, 2021). As social media platforms are fundamental to brands' and organizations' marketing activities, the potential impact of fake news for both companies and consumers is substantial (Di Domenico et al., 2021; Zaman et al., 2023). Fake news can negatively influence consumers' attitudes toward brands (Flostrand et al., 2019), instilling misleading beliefs and distrust, and curbing consumption (Visentin et al., 2019). Fake news reports can unleash public relations nightmares, which ultimately have a negative financial effect on business sales and stock prices (Atkinson, 2019).

Past research has identified several linguistic features that differentiate fake news from real news, using automated text analysis to quantify and examine these features (e.g., Asubiaro & Rubin, 2018; Gupta et al., 2022; Damstra et al., 2021; Singh, Ghosh, & Sonagara, 2021; Di Domenico et al., 2021). However, to our knowledge, no work has applied narrative theory to characterize fake (relative to real) news. This is surprising, given that guides to identifying misinformation anecdotally note fake news' tendency to adopt a sensationalist story format

(Ireton & Posetti, 2018; PBS, 2021) coupled with media content's broader use of narrative structure (Roeh, 1989; Thorsen & Jackson, 2018).

A wealth of business research has productively applied narrative theory to understand the story-based aspects that predict consumers' responses to marketing stimuli such as advertisements (Escalas, 2007; Lien & Chen, 2013), brand biographies (Paharia et al., 2011), and crowdfunding pitches (Moradi & Badrinarayanan, 2021). More recently, research has coupled this theoretical lens with automated text analysis to predict consumer response in contexts such as consumer reviews (van Laer et al., 2019) and movies (Berger, Kim, & Meyer, 2021). We contribute to the growing literature on narrative persuasion that leverages automated text analysis (Berger et al., 2021; Moore & McFerran, 2017; van Laer et al., 2019) by examining narrativity in the context of fake (versus real) news. Specifically, we examine whether real and fake news differ in terms of their narrative content (such as sentiment volatility, period-to-period changes over the course of the article; Berger et al., 2021) and discourse (such as story shape, the rise and fall in sentiment throughout a story, based on the plot; van Laer et al., 2019), among other narrativity elements.

We contribute in two ways to the growing literature on word-of-mouth and fake news both within and outside of marketing: While past work has examined how real and fake news differ in linguistic dimensions (Asubiaro & Rubin, 2018; Damstra et al., 2021; Di Domenico et al., 2021), we are the first (to our knowledge) to integrate literature from the field of narrative persuasion to examine how real vs. fake news differ in terms of their narrativity. Second, we examine how these narrativity elements predict content sharing, an outcome that generates business value (Babić Rosario et al., 2016).

Relatedly, we contribute to the literature on narrative persuasion: we examine whether the narrativity elements examined in prior work influence message content sharing, which is distinct from the primarily attitude-based outcomes examined in past work on narrative persuasion (Berger et al., 2021; van Laer et al., 2019). Our results replicate some past findings using attitude as an outcome but not others, suggesting that there may be boundary conditions that can be examined with follow-up research.

2. Conceptual Development

2.1 Language in News and Narrative Persuasion

Research aiming to distinguish fake news (articles that are intentionally and verifiably false; Ireton & Posetti, 2018) and real news has shown that they differ in their linguistic characteristics: fake news reports tend to be shorter than true news stories, with fewer and shorter words and sentences (Asubiaro & Rubin, 2018; Zhou et al., 2020), and use excessive capitalization and punctuation (Bradshaw et al., 2020; Marchal et al., 2019). Fake news is more likely to use affective and particularly negative language (Damstra et al., 2021), featuring more fear-, anger-, and sadness-related words than real news (Gupta et al., 2022). Fake news is also more likely to feature first- and second-person pronouns (Di Domenico et al., 2021; Singh, Ghosh, & Sonagara, 2021), to feature more novel language than regular news (Ruffo et al., 2023; Vosoughi et al., 2018) and is simpler to process than real news articles (Ruffo et al., 2023).

These differences presumably stem from fake news writers' motives to create articles (Di Domenico et al., 2021). Their primary motivation is financial: fake news may be created to encourage clicks and revenues from advertising. Another motivation is political, in which the

writer aims to influence public perception about a political candidate or topic (Plangger & Campbell, 2022). Through the strategic presentation of their content, fake news creators attempt to make their content appear credible (Di Domenico et al., 2021). Writers aim to attract and sustain readers' attention to enable its spread through social media (Zhang & Ghorbani, 2020).

Given fake news authors' motives to fabricate legitimacy and attract attention, it is perhaps not surprising that fake news prominently features a story format (Tamul et al., 2020). Narratives (or stories) are a message format depicting actors pursuing goals over time (Bruner, 1986). By this definition, nearly all news articles adopt a narrative format. Although past work has distinguished between "hard" news (usually regarding politics or economics) and "soft" news (e.g., gossip or human interest stories), these articles usually describe some type of event as well as the people involved (Reinemann et al., 2012). News articles can vary in their extent of narrativity – or, the extent to which they tell a story -- however. Narrative journalism has become prevalent in recent years (Van Krieken & Sanders, 2021; Wardle, 2006), particularly on social media platforms (Casero-Ripollés et al., 2020). Narrativity in news articles has become increasingly influential in recent years, as well: some have argued that the contemporary 'post-truth' media context, in which objective facts are less important than appeals to emotion and personal belief, prioritizes narrativity (Berthon & Pitt, 2018; McDermott, 2019).

Prior research on storytelling shows that stories as a message format are particularly likely to achieve fake news writers' goals of enhancing credibility, gaining and sustaining attention, influencing beliefs, and attaining virality. Messages crafted in narrative form may have a retransmission advantage because stories are a fundamental form of human cognition and communication and are easier to comprehend and recall (Bruner, 1986). To evaluate whether something is true, people attend to whether something is easy to process (Schwarz & Jalbert,

2020). Stories also deliver information in a vivid and engaging manner; readers may become transported into the story, a form of processing involving enhanced attention, emotional engagement, and generation of imagery (Green & Brock, 2000). Finally, narratives also encourage mental simulation of events (Escalas, 2004), which may make them seem more realistic and authentic (Plangger & Campbell, 2022).

In the current work, we move beyond the textual features examined by past research on fake news to consider how narrative content and discourse characteristics typify real and fake news, and how these characteristics are associated with the article's likelihood of being shared.

2.2 Narrative Content and Discourse

While narratives are generally more engaging than didactic message formats (Oschatz & Marker, 2020), narrative messages vary in ways that impact their persuasiveness. In particular, narratologists distinguish between narrative content (what is conveyed) and discourse (how it is conveyed; Fludernik, 2009). Recent work in marketing and consumer research has applied these distinctions to understand how consumers respond to marketing and media content. For instance, Van Laer et al. (2019) examined how several prominent narrative content and discourse characteristics determined the persuasiveness of consumer reviews. In terms of narrative content, they examined the effects of characters' landscapes of affective consciousness (the extent to which a text recounts initial events about which characters express feelings that lead to subsequent events; for instance, "I felt nervous, so I took a few deep breaths"; Bruner, 1986) and cognitive consciousness (the extent to which a character recounts initial events about which characters express thoughts that, in turn, lead to subsequent events; for instance, "I thought I was

going to be late, so I sped up my pace.”). In addition, they examined how a story’s spatial embedding (the extent to which a text focuses on particular spaces and names its attributes, rather than develops categorizations or generalizations) and temporal embedding (the chronological flow of the events indicating the direction of the story) influence story persuasiveness. They found that each feature (affective and cognitive consciousness, as well as temporal and spatial embedding) had a positive relationship with attitude (number of “likes” associated with the featured review).

Separately, Berger, Kim, & Meyer (2021) introduced the construct of sentiment volatility, “period-to-period shifts in sentiment” (p. 236), and examined how it is associated with consumers’ liking of movies and news articles. We consider sentiment volatility to be a content characteristic of narratives, as plots vary in their frequencies of narrative transitions. Berger et al. (2021) demonstrate that sentiment volatility makes experiences more stimulating and engaging, drawing on past work suggesting that content variety reduces boredom (Ratner, Kahn, & Kahneman, 1999) and can make an experience feel more exhilarating (Etkin & Mogilner, 2016). People experience this increase in engagement favorably, which enhances their attitude toward the focal content.

In terms of narrative discourse, van Laer et al. (2019) examined the role of story shape and story drama. They defined story shape (or genre) in terms of Gergen & Gergen’s (1988) classic taxonomy, which includes five categories: In a stable story shape, the events neither improve nor decline over time with respect to a character’s goals. In a progressive story shape, events continuously improve for characters throughout the story line, whereas in a regressive shape, they decline. A comedy features a U-shaped storyline, in which events become worse and then better for a character. A tragedy features an inverted U shape, in which events improve and

then decline with respect to the character's goals. See figure 1 for a depiction of these story shapes. Research on fictional and personal stories has corroborated the presence of these shapes in stories (Fivush, Thorstand, & Graci, 2021; Reagan et al., 2016). Van Laer et al. (2019) found that stories with complex shapes (comedies and tragedies) enhanced positive review feedback (i.e., liking) relative to stories with stable shapes.

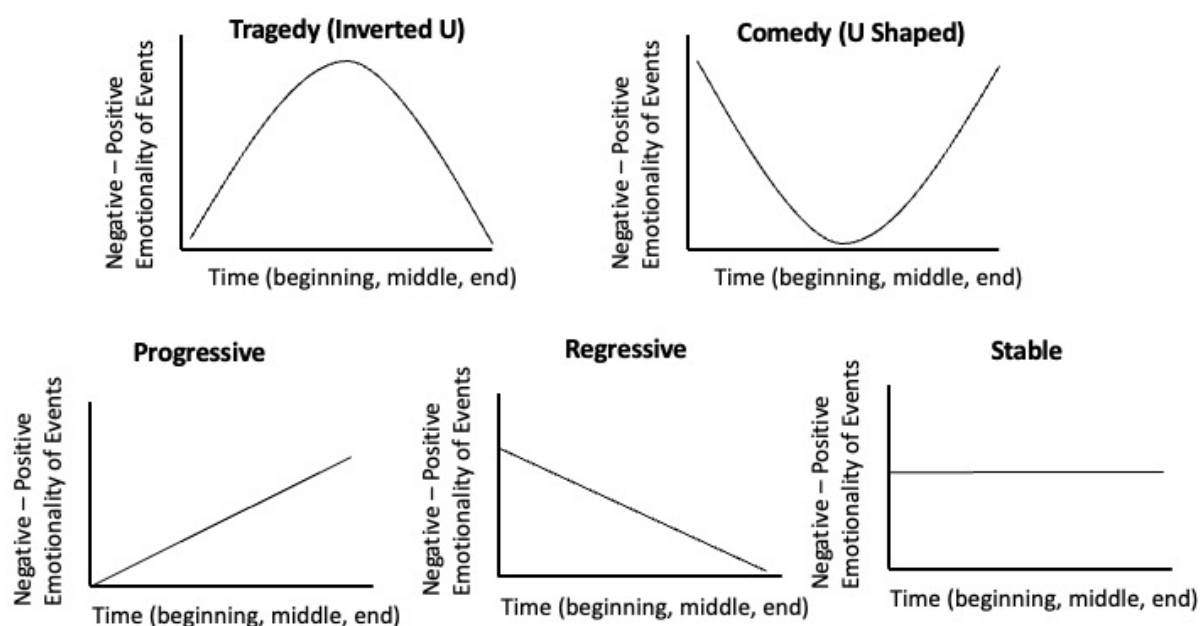


Fig. 1. Five common story shapes

Furthermore, the plot points of a story can be edited and arranged so that the emotional climax occurs earlier or later in a story – that is, the type of story drama, or the order of story events. Specifically, authors can present their stories in a curiosity order (in which an emotional climax occurs earlier in a text) or a surprise order (in which the emotional climax occurs later in the story). Because a curiosity order engages readers with the narrative world to understand the

origin of the emotional introduction, van Laer et al. (2019) predicted and found that an earlier emotional climax (curiosity order) was associated with higher review persuasiveness.

2.3 Development of Predictions

In the current work, we examine how real and fake news differ in terms of narrative content and discourse characteristics, contributing to the growing literature on the ways that real and fake news differ in terms of linguistic characteristics (Asubiaro & Rubin, 2018; Damstra et al., 2021; Di Domenico et al., 2021). We also contribute to the literature on narrative persuasion beyond van Laer et al. (2019) and Berger et al. (2021), who examined the effects of narrative features on attitude-related outcomes in the context of real or overtly fictional messages. In contrast, the current work examines whether certain types of messages (e.g., real or fake news) differ systematically in terms of their prevalence of narrative features, as well as in their effects on media sharing behavior. See table 1 for an overview of past findings and our predictions.

We draw from the deception detection and reality monitoring literatures, as well as prior research on fake news, to guide our predictions. According to this research, the content of stories based on true versus fake experiences differs because of the processes underlying their generation (Johnson & Raye, 1981; Nahari, 2018; Sporer, 1997). Real experiences are encoded through perceptual processes and are thus more likely to contain vivid sensory information (details of smell, taste, or touch), visual, spatial, and temporal details, and information about the character's emotional experience when recounted. In contrast, stories about fabricated events are derived from an internal source and are more likely to contain cognitive operations, such as thoughts and reasonings, and are usually vague and less concrete when they are told (Sporer,

1997). Separately, research on storytelling has shown that when people tell stories to entertain (versus recount events accurately) – which may characterize fake news writers’ motives – they are less likely to include sensory information and more likely to include opinions and evaluations (Dudukovic et al., 2004). Taken together, we predict that real news articles should be more likely to feature a higher degree of temporal and spatial embedding.

We also predict that real news will feature a higher degree of affective and cognitive consciousness. While past research on fake news has shown a higher prevalence of emotions in fake content (Damstra et al., 2021; Gupta et al., 2022), affective consciousness as defined by van Laer et al. (2019) and Bruner (1986) focuses specifically on the extent to which a character expresses feelings about initial events that lead to subsequent events, which is distinct from the presence of mere positive and negative sentiment in a text. Similarly, the landscape of cognitive consciousness involves characters’ expressions of thoughts that, in turn, lead to subsequent events. Thus, both constructs focus on explaining the reasons behind events and fostering understanding, which is more likely to typify the motives of real news writers (Graves, Nyhan, & Reifler 2016).

However, because of fake news’ broader association with higher arousal emotions (Damstra et al., 2021; Zhang et al., 2021), writers’ motives to attract and sustain attention (Zhang & Ghorbani, 2020), and noted sensationalist content (Ireton & Posetti, 2018; PBS, 2021), we expect fake (vs. real) news to be higher in sentiment volatility. Formally, we predict the following:

H1: Real (vs. fake) news features a higher degree of (a) temporal embedding, (b) spatial embedding, (c) affective consciousness, and (d) cognitive consciousness, while fake (vs. real) news features a higher degree of (e) sentiment volatility.

We also examine the prevalence of story shape and story drama (curiosity vs. suspense ordering) in real and fake news. On the one hand, prior research on fake news suggests that fake news is more likely to feature novelty than real news (Ruffo et al., 2023; Vosoughi et al., 2018), which may suggest that fake news is more likely to adopt a story shape with more variation (corresponding with comedy and tragedy shapes). Alternatively, fake news may be more likely to feature an overall negative trend (corresponding with tragedy and regressive shapes) because of fake news writers' motives to attract attention, which is more likely to be achieved with negative language (Damstra et al., 2021; Gupta et al., 2022). Finally, because past research suggests that fake news articles are simpler, they may be more likely to feature stable shapes (Ruffo et al., 2023). Given the range of possible predictions and the lack of a strong a priori basis for predicting that real and fake news differ in terms of their overall story shape and curiosity versus suspense order, we pose the following research question:

RQ1: Do real and fake news differ in terms of their (a) progressive, (b) regressive, (c) comedy, (d) tragedy, (e) and stable shapes, as well as their (f) drama ordering?

Van Laer et al. (2019) and Berger et al. (2021) examined content liking or attitude as their outcome measure of persuasion. In the current work, we examine news article sharing as our outcome measure of interest, thus contributing beyond van Laer et al. (2019) and Berger et

al. (2021). In general, the more people like a story, the more likely they should be to share that content (Quesenberry & Coolsen, 2019). However, research on word-of-mouth shows that sharing behavior differs from liking in several ways and that people share content for three main reasons. The first motivation relates to impression management (Babić Rosario, De Valck, & Sotgiu, 2020); for instance, people may believe that sharing interesting and engaging information with others may confer those qualities upon themselves (“because I share interesting content, people view me as interesting”). The second motivation is social: people share information to engage with their community and feel part of a group (Oh & Syn, 2015). Relatedly, people may share or comment on content to obtain visibility in a public space; they share content as a way to foster self-publicity (Arvidsson & Caliendo, 2016). Finally, people share news for altruistic reasons to try to help others (Lovett, Peres, & Shachar, 2013).

A separate stream of research examines aspects of message content that enhance sharing behavior. Past work has repeatedly shown that emotional, arousing content is more likely to be shared (Berger & Milkman, 2012; Vosoughi et al., 2018). Additionally, Rimé’s (2009) influential theory of emotion regulation suggests that the experience of emotional arousal prompts social sharing of that emotion because it helps people make sense of the emotional experience and strengthens social bonds. As previously outlined, narratives are particularly effective in arousing emotions relative to other message formats (van Laer et al., 2014).

Narratives encourage mental simulation of events (Escalas, 2004), which may make them seem more realistic and authentic. Past research on news article sharing has shown that the level of belief that the articles were accurate and truthful was the strongest predictor of likelihood of sharing (Buchanan, 2020; Di Dominico et al., 2021; Koohikamali & Sidorova, 2017) versus other factors such as how many people had engaged with it and the source of the message.

However, some research has shown that social media sharing behavior can actually be unrelated to persuasion or judgments about the article's accuracy (Pennycook & Rand, 2021). Social media content may distract people from prioritizing truth when they decide what to share, favoring aspects that they perceive as more likely to engage others. For instance, Chen, Pennycook and Rand (2021) found that message evocativeness predicted message sharing intentions, and narratives tend to be more vivid and imagery-inducing than their nonnarrative counterparts (van Laer et al., 2014).

Collectively, prior work on word-of-mouth shows that articles that arouse emotion, seem truthful, and are evocative are more likely to be shared with others. Separately, research has shown that narrative format messages are perceived as more emotional, truthful, and entertaining than their nonnarrative counterparts. We extend this logic to predict that news articles that are higher in narrativity (i.e., feature narrative content and discourse elements more prevalently) will enhance sharing. Formally:

H2: Content-sharing behavior is associated with a higher degree of (a) temporal embedding, (b) spatial embedding, (c) affective consciousness, (d) cognitive consciousness, (e) sentiment volatility, (f) drama (curiosity order), (g) progressive, (h) regressive, (i), comedy and (j) tragedy shapes.

We test these hypotheses in 3 studies. The first study examines these hypotheses in the context of real and fake news secondary datasets that focus on political and health-related news, using actual consumer tweets (shares) as the outcome variable. The second study uses a subset of the articles from study 1 to examine the robustness of the results when participants indicate their

sharing intentions. The third study examines the differences between real and fake news in terms of narrativity and their association with sharing in the context of another secondary dataset focused on celebrity gossip. See table 1 for an overview of the predictions and findings.

Table 1. Overview of Past Literature, Predictions, and Findings

Construct	Definition	Relevant Literature	Prediction in the Current Work	Comparing real/fake news		Effect on shares		
Content				Study 1	Study 3	Study 1	Study 2	Study 3
Temporal Embedding (TE)	Chronological flow and causal links between events	van Laer et al. (2019) found TE enhances review liking	Real news is higher in TE; TE increases sharing	R > F	R < F	n.s.	n.s.	n.s.
Spatial Embedding (SE)	Focuses on particular spaces	van Laer et al. (2019) found that SE enhances review liking	Real news is higher in SE; SE increases sharing	R > F	R > F	n.s.	n.s.	n.s.
Landscape of Affective Consciousness (LoA)	Characters express feelings about initial events that lead to subsequent events	van Laer et al. (2019) found LoA enhances review liking	Real news is higher in LoA; LoA increases sharing	R > F	n.s.	decreases shares	decreases sharing intentions	decreases shares
Landscape of Cognitive Consciousness (LoC)	Characters express thoughts about initial events that lead to subsequent events	van Laer et al. (2019) found LoC enhances review liking	Real news is higher in LoC; LoC increases sharing	R > F	n.s.	increases shares	increases sharing intentions	decreases shares
Sentiment Volatility (SV)	Period-to-period shifts in sentiment	Berger et al. (2021) found SV enhances attitude toward and engagement in content	Fake news is higher in SV; SV increases sharing	R < F	R < F	decreases shares	decreases sharing intentions	decreases shares
Discourse								
Progressive Genre	Emotion improves over the course of a text	van Laer et al. (2019) found no effect of progressive (vs. stable) shapes on review liking	Progressive increases sharing relative to stable	R < F	n.s.	decreases shares	n.s.	n.s.
Regressive Genre	Emotion deteriorates over the course of a text	van Laer et al. (2019) found a negative effect of regressive (vs. stable) shapes on liking	Regressive increases sharing relative to stable	n.s.	R > F	n.s.	n.s.	n.s.
Comedy Genre (U-shaped)	Emotion first deteriorates and then ameliorates over the text	van Laer et al. (2019) found a positive effect of comedy (vs. stable) shapes on review liking	Comedy increases sharing relative to stable	R < F	n.s.	decreases shares	n.s.	n.s.
Tragedy Genre (Inverted U)	Emotion first ameliorates and then deteriorates over the text	van Laer et al. (2019) found a positive effect of comedy (vs. stable) shapes on review liking	Tragedy increases sharing relative to stable	n.s.	n.s.	increases shares	increases sharing intentions	increases shares
Stable Genre	Emotion is stable over the text	van laer et al. (2019) treated "stable" as the reference group		R > F	n.s.	stable genre treated as reference group		
Curiosity-order Drama	The emotional climax is located earlier (vs. later) in a text	van Laer et al. (2019) found a positive effect of comedy order dramas on review liking	Curiosity-order drama enhances sharing	n.s.	n.s.	n.s.	n.s.	n.s.

NOTE: Bold findings indicate convergence across studies

3. Study 1: Examining the Role of Narrativity in Real vs. Fake News Datasets

We computed narrative features from existing news articles and combined them with additional sources of information to test our hypotheses. Specifically, we obtained several datasets of verified fake and real news and the number of shares on Twitter for each of these articles.

Twitter is a social media site whose influence has been compellingly demonstrated in the context of politics and in shaping consumer behavior (Buccoliero et al., 2016; Hennig-Thurau et al., 2015). We followed the methodology outlined in van Laer et al. (2019) and Berger et al. (2021) to operationalize our focal narrative constructs: we used text analysis software (the Linguistic Inquiry and Word Count program; Pennebaker et al., 2015) to extract key features, which were used to compute the focal narrativity features. In addition to examining whether the focal narrativity features are associated with news article sharing (H2), we examine whether real and fake news differ systematically in terms of narrativity (H1, RQ1).

3.1 Method

3.1.1 Sampling Frame. There is generally limited availability of large-scale benchmarks for fake news detection, especially where the truth labels are assigned via fact-checking, which is a time-consuming activity. We used three preexisting datasets compiled by computer science researchers, which all contain the number of times the article was shared on Twitter:

The ReCOVery dataset contains news items regarding COVID-19. ReCOVery contains the news article content and the number of times the article was shared on Twitter. News labels (fake or real) were assigned according to the reliability of the news publisher, which was retrieved from two data sources, namely, NewsGuard ([newsguardtech.com](https://www.newsguardtech.com)) and Media Bias/Fact

Check (mediabiasfactcheck.com) websites. The ReCOVery dataset contains 2,017 news articles, 663 fake and 1,354 real. More information about this dataset can be found in Zhou et al. (2020b).

The second dataset contained news related to politics and categorized articles based on labels collected from the fact-checking website PolitiFact. PolitiFact features truth labels on a continuous 6-point scale ranging from “pants on fire” (“The statement is not accurate and makes a ridiculous claim”) to “true” (“The statement is accurate and there’s nothing significant missing”). This dataset did not include articles labeled mostly false and half true, given their mixed truth status. After cleaning missing news articles from the dataset, we obtained a total of 853 news articles related to politics, 603 fake and 250 real. In addition, this dataset contains the number of times the article was shared on Twitter. More information about this dataset can be found in Shu et al. (2020; see also Shu, 2019).

The final dataset is similar to the second dataset but was collected by separate authors at a different time period (Shrestha et al., 2023): it contains 1,171 politically related news items that were also fact-checked via the PolitiFact website and that were clearly “true” or “false” (i.e., no mostly false or half true status). In addition, this dataset contains the number of times the article was shared on Twitter. This dataset contains a total of 936 articles: 417 total fake news articles and 519 real news articles. See Web Appendix A for additional information about the datasets.

We combined our three datasets into a master dataset (and retained a variable indicating from which specific dataset the news article originated). We retained articles that were 7 sentences or longer, following van Laer et al. (2019). Our final dataset consisted of 2,892 news articles: 1,699 real and 1,193 fake. The articles averaged 45.87 sentences ($SD = 62.69$), 897.74 words ($SD = 1,032.58$) and 136.61 shares ($SD = 882.46$).

3.1.2 Operationalization of Narrativity. Following past research, we used the Linguistic Inquiry and Word Count (LIWC) program (Pennebaker et al., 2015) to derive and compute the narrative features of interest and control variables. LIWC compares each word in a text against predefined word categories, classified in dictionaries. It then identifies the number of words that match each dictionary to determine an intensity score per dictionary. For example, if LIWC analyzed a document containing 1,000 words, it might find that 70 words are related to negative emotions and 20 words are related to affiliation. LIWC would convert these numbers to percentages: 7.0% negative emotion and 2.0% affiliation. Marketing scholars have used LIWC summary dictionaries to forecast social media engagement (Hwong et al., 2017) or examine how the emotional style of users' Facebook posts influences company revenue (Yoon et al., 2018). More germane to our present context, authors have used sentiment analysis dictionaries to examine narrativity's association with persuasion in consumer reviews (van Laer et al., 2019), celebrities' uses of narratives in social media (Aleti et al., 2019) and how sentiment volatility is associated with cultural success (Berger et al., 2021).

We use van Laer et al.'s (2019) methodology to compute landscapes of consciousness, landscape of action, spatial and temporal embedding, the five story shapes, and drama (curiosity versus suspense order). We also used Berger et al.'s (2021) method for computing sentiment volatility; in brief, we first identified the average sentiment for each sentence in a news article and then calculated the standard deviation of the differences in sentiment between contiguous sentences. See Web Appendix B for detailed calculation descriptions.

We also measured and controlled for a number of other text features that past work has shown are associated with consumers' responses to messages on social media. Aleti et al. (2019) use LIWC's analytic dictionary to measure a social media post's degree of analytical style

(ranging from more analytical, exemplified by using more prepositions and references to complexly organized objects, to less analytical, exemplified by using more personal pronouns and auxiliary verbs; see Pennebaker et al., 2015), the clout dictionary to measure focalization (ranging from a more internal focus, exemplified by using first-person singular pronouns, versus a more external focus, exemplified by using more second-person singular pronouns), and the tone dictionary to measure a post's emotional style (more negative to more positive emotional style). Finally, to control for length, we include the number of sentences as well as the overall word count and words per sentence (given past work showing that fake news reports tend to use fewer and shorter words and sentences; Asubiaro & Rubin, 2018; Zhou et al., 2020a).

3.2 Results

3.2.1 Comparing Real and Fake News. To test hypothesis 1, we conducted a series of ANCOVAs to compare real and fake news on the dimensions of interest. We included word count and number of sentences in all analyses to control for article length, as well as the specific dataset from which the article originated. Table 1 reports the full set of results, that is, the mean prevalence of the narrativity in real and fake news. To test RQ1, we created dummy variables denoting the story shape and used this measure in a logistic regression, including article reliability (real or fake) as well as the length and dataset controls mentioned above.

Overall, we found that real (vs. fake) news contained a higher degree of temporal embedding and spatial embedding and landscapes of affective and cognitive consciousness (hypotheses 1a - 1d were supported). Fake news contained a higher degree of sentiment volatility, supporting hypothesis 1e.

Addressing RQ1, there were significantly more progressive and marginally more comedy shapes in fake news than in real news and more stable shapes in real news. There were no

differences between real and fake news in terms of whether they conformed to regressive or tragedy shapes or whether the climax occurred earlier in the article (i.e., curiosity order). Fake news was higher in clout (a more external focus) and more negative in tone. See table 2.¹

Table 2. Comparing Real versus Fake News in Terms of Narrative Features, Study 1

	Mean (SD) Fake News/count (% overall total)	Mean (SD) Real News count (% overall total)	<i>F</i> test/ <i>b</i> coefficient in logistic regression	Hypothesis test/Research question
Temporal embedding	1.97 (.12)	1.99 (.20)	$F(1,2887) = 14.19, p < .001, \eta_p^2 = .005$	H1a, supported
Spatial embedding	1.98 (.14)	1.99 (.09)	$F(1,2887) = 5.71, p = .02, \eta_p^2 = .002$	H1b, supported
Landscape of affective consciousness.	.002 (.01)	.003 (.01)	$F(1,2887) = 17.95, p < .001, \eta_p^2 = .006$	H1c, supported
Landscape of cognitive consciousness	.001 (.01)	.002 (.01)	$F(1,2887) = 4.44, p = .04, \eta_p^2 = .002$	H1d, supported
Sentiment volatility	.21 (.08)	.19 (.07)	$F(1,2887) = 89.67, p < .001, \eta_p^2 = .03$	H1e, supported
Progressive	80 (2.8%)	92 (3.2%)	$b = -.39, p = .04, \text{Wald} = 4.14$	RQ1a
Regressive	29 (1.0%)	49 (1.7%)	$b = .11, p = .69, \text{Wald} = .16$	RQ1b
Comedy (U shaped)	55 (1.9%)	85 (2.9%)	$b = -.36, p = .09, \text{Wald} = 2.86$	RQ1c
Tragedy (Inverted U)	35 (1.2%)	47 (1.6%)	$b = -.10, p = .73, \text{Wald} = .12$	RQ1d
Stable	992 (34.3%)	1,426 (49.3%)	$b = .32, p = .01, \text{Wald} = 6.65$	RQ1d
Curiosity order drama	.60 (.29)	.57 (.28)	$F(1,2887) = 2.53, p = .11$	RQ1f
Analytic	89.12 (11.15)	87.50 (11.80)	$F(1,2887) = .54, p = .46$	
Clout	69.75 (12.66)	68.87(12.00)	$F(1,2887) = 7.66, p < .01, \eta_p^2 = .03$	
Tone	35.79 (22.78)	40.37 (22.04)	$F(1,2887) = 26.65, p < .001, \eta_p^2 = .01$	

3.2.2 Predicting Shares Based on Narrativity. We used the number of times the news article was shared as the dependent variable in our model. Following past work, to account for the skewed distribution of tweets (as is common with Twitter data), we used a negative binomial regression analysis (Ahmadi, Waltenrath, & Janze, 2022; Jalali & Papatla, 2019; Pezzuti, Leonhardt, & Warren, 2021).

¹ The mean values for the narrativity dimensions are a function of how they are calculated and do not have a readily-translatable metric, like a “count.” These values are in line with those obtained by van Laer et al. (2019). Much like a value of 2 on a 7-point likert scale does not have objective meaning, these values become meaningful when compared with other values that were derived in the same way.

We report the effects of the narrative elements on content sharing in table 3. The first model consists of the control variables. In the second model, we entered the focal narrativity elements (omitting the stable shape dummy variable, which serves as the reference group category). This model demonstrates a significant improvement in fit ($\Delta\chi^2(10) = 96.93, p < .001$; $\Delta AIC = 12.14$). The likelihood ratio test for the negative binomial regression model was significant ($\chi^2(34) = 1038.92, p < .001$), indicating good fit. A higher degree of landscape of cognitive consciousness and tragedy shapes increased shares (hypotheses 2d and were 2h supported), but landscape of affective consciousness, sentiment volatility, and progressive and comedy shapes significantly *decreased* shares. The effects of spatial and temporal embedding and curiosity drama order were not significant. Real news was also shared more than fake news, as was more analytic and more negative news.

Table 3. Predicting Sharing Intentions Based on Narrative Features, Study 1.

Parameter	Model 1: Controls			Model 2: Main effects			Hypothesis Test
	B	SE	p	B	SE	p	
(Intercept)	5.50	.27	.00	5.11	.56	.00	
Word count	.00	.00	.64	.00	.00	.63	
Number of sentences	.00	.00	.05	.01	.01	.02	
Reliability (1 = real, 0 = fake)	.52	.05	.00	.51	.05	.00	
Analytic	.01	.00	.00	.04	.01	.04	
Clout	.00	.00	.43	.00	.01	.80	
Tone	-.01	.00	.00	-.01	.01	.00	
Dataset 1	-1.60	.10	.00	-1.57	.10	.00	
Dataset 2	-1.26	.07	.00	-1.23	.08	.00	
Article date	.00	.00	.00	.00	.13	.00	
Temporal embedding				.06	.21	.24	Hypothesis 2a not supported
Spatial embedding				.34	.29	.77	Hypothesis 2b not supported
Landscape of affective consciousness				-31.02	10.17	.00	Hypothesis 2c not supported
Landscape of cognitive consciousness				29.85	15.32	.05	Hypothesis 2d supported
Sentiment volatility				-.79	.36	.03	Hypothesis 2e not supported
Curiosity drama order				.06	.07	.42	Hypothesis 2f not supported
Progressive				-.28	.09	.00	Hypothesis 2g not supported

Regressive	.16	.13	.22	Hypothesis 2h not supported
Comedy (U shaped)	-.63	.10	.00	Hypothesis 2i not supported
Tragedy shape (Inverted U)	.73	.13	.00	Hypothesis 2j supported
Pearson X^2	7,547.45 (2,312)	7,450.52 (2302); X^2 Change(10) = 96.93, $p < .001$		
AIC	24,602.77	24,590.63; AIC change = 12.14		

3.3 Discussion

Study 1 found support for hypotheses 1a-1e (real news is more likely to feature temporal and spatial embedding and content describing landscapes of affective and cognitive consciousness; fake news is more likely to feature sentiment volatility) and further found that fake news is more likely to feature progressive and comedy shapes and less likely to feature stable shapes. Study 1 also provided support for hypotheses 2d and 2j; the landscape of cognitive consciousness and tragedy shapes increase shares, but the landscape of affective consciousness, sentiment volatility, and progressive and comedy shapes *decrease* shares. One of the limitations of study 1 is associated with the outcome of “shares” as captured in the secondary datasets from Twitter: Twitter has undergone a number of changes over the years that could impact participants’ likelihood of sharing the news article. While we controlled for publication date in study 1, study 2 addresses the concern by asking participants to indicate their sharing intentions directly.

4. Study 2: Replicating Effects with Lab Participants

In study 2, research participants evaluated a subsample of study 1’s articles to address hypothesis 2 and RQ1. We then examined whether the narrativity elements associated with the articles are related to participants’ sharing intentions.

4.1 Method

U.S. based Prolific Academic participants ($N = 302$, $M_{\text{age}} = 34.69$, 57.7% female) rated articles drawn from study 1's dataset. Participants were presented with 10 articles for evaluation. These articles were randomly sampled from a pool of 100 articles from study 1. This pool contained articles that were between 22 and 30 sentences (or 26 sentences ± 4); that is, they are long enough to be representative of the types of news articles shared online but short enough to reduce the burden on the participant. After reading each article, participants rated the extent to which they would be willing to share and like each news article on social media (on 1 = not at all to 7 = a great deal scales; we used a two-item outcome measure given the reduced reliability and information provided by single item measure; Bergkvist & Rossiter, 2007). These responses were highly correlated (ranging between $r = .75$ and $.95$ per article, $p < .001$) and were combined into a composite measure of sharing intentions.

4.2 Results

We conducted a regression analysis to examine whether the effect of the narrative elements examined in study 1 predicts measured sharing intentions. The R^2 of the model including the experimental values increased significantly from .10 (controls-only model) to .38 (R^2 change = .28, $p < .01$). The landscape of cognitive consciousness and tragedy shape (inverted U) increased shares, supporting hypotheses 2e and 2j; no other hypotheses were supported. Replicating the findings from study 1, a higher degree of landscape of affective consciousness and sentiment volatility decreased shares. We discuss these findings, and how they correspond with the findings of the prior study, in the general discussion. See table 4.

Table 4. Predicting Sharing Intentions Based on Narrative Features, Study 2.

Parameter	Model 1: Controls			Model 2: Main effects			Hypothesis Test
	B	SE	<i>p</i>	B	SE	<i>p</i>	
(Intercept)	4.89	1.86	.01	4.22	1.96	.03	

Word count	.00	.00	.64	.00	.00	.60	
Number of sentences	.01	.02	.57	.00	.02	.96	
Reliability (1 = real, 0 = fake)	.05	.11	.64	-.03	.11	.82	
Analytic	-.01	.01	.23	-.01	.01	.36	
Clout	.00	.01	.89	.00	.00	.71	
Tone	.00	.00	.06	.00	.00	.19	
Article date	.00	.00	.38	.00	.00	.89	
News article	.00	.00	.91	-.01	.01	.04	
Temporal embedding				-.22	.18	.23	Hypothesis 2a not supported
Spatial embedding				.12	.48	.80	Hypothesis 2b not supported
Landscape of affective consciousness				-55.80	24.44	.03	Hypothesis 2c not supported
Landscape of cognitive consciousness				82.13	28.21	.01	Hypothesis 2d supported
Sentiment volatility				-1.86	.87	.04	Hypothesis 2e not supported
Curiosity drama order				-.10	.19	.60	Hypothesis 2f not supported
Progressive				.27	.24	.27	Hypothesis 2g not supported
Regressive				.35	.33	.29	Hypothesis 2h not supported
Comedy (U shaped)				.30	.25	.24	Hypothesis 2i not supported
Tragedy shape (Inverted U)				.83	.39	.04	Hypothesis 2j supported

4.3 Discussion

This study replicated some of the findings from study 1 (which used the number of shares on Twitter as the outcome variable) when study participants indicated their sharing intentions: participants were more likely to share articles with a higher landscape of cognitive consciousness and tragedy shape. Contrary to the hypotheses, but replicating study 1, participants were *less* likely to share news articles that contained content higher in landscape of affective consciousness and sentiment volatility. In the next study, we examine the replicability of these findings in another substantive context. While health and political news – the focus of the datasets used in studies 1 and 2 – are relevant to consumers and the domain of marketing (Kotler & Levy, 1969), study 3 examines news article narrativity in the more traditional marketing context of person brands or celebrities.

5. Study 3: Replicating Effects in a Celebrity-Focused Dataset

Study 3 examines whether the effects from studies 1 and 2 are replicated in a dataset associated with person brands or celebrities (Centeno & Wang, 2017). This study provides evidence of robustness across contexts.

5.1 Method

We obtained an additional dataset (Shu, 2019) that contained articles identified as real and fake. The fake news articles originated from GossipCop, a website for fact-checking entertainment stories aggregated from various media outlets. The real news articles originated from E! Online, which is a well-known media website for publishing entertainment news pieces. See Web Appendix B for additional information about how the true and fake news statuses were determined. This dataset also contained information about the number of times the news article was shared on Twitter.

As in study 1, and following van Laer et al. (2019), we retained articles that were 7 sentences or longer. Our final dataset consisted of 22,205 news articles: 16,980 real and 5,225 fake. The articles averaged 35.31 sentences ($SD = 61.57$), 745.30 words ($SD = 1,284.47$) and 126.55 shares ($SD = 263.48$). We computed the same variables as those derived for study 1 and conducted the same analyses.

5.2 Results and Discussion

5.2.1 Comparing Real and Fake News. To test hypothesis 1, we again conducted a series of ANCOVAs to compare real and fake news on the dimensions of interest, including word count and number of sentences in all analyses to control for article length. Table 5 reports the full set of results. To test RQ1, we again created dummy variables denoting the story shape and used this measure in a logistic regression, including article reliability (real or fake) as well as the article length controls mentioned above.

Overall, we found that real news contained a higher degree of spatial embedding and fake news contained more sentiment volatility (hypotheses 1b and 1e were supported). There were no differences in any of the other narrativity elements. Two of the hypothesized findings from study 1 in terms of comparing real and fake news were thus replicated across datasets. We consider this finding in the discussion. In addition, real news was more analytic, lower in clout, and more positive in tone. See table 5.

Table 5. Comparing Real versus Fake News in Terms of Narrative Features, Study 3

	Mean (SD) Fake News / count (% overall)	Mean (SD) Real News count (% overall)	<i>F</i> test / <i>b</i> coefficient in logistic regression	Hypothesis test / Research question
Temporal embedding	1.96 (.21)	1.94 (.25)	$F(1,22,200) = 44.57, p < .001, \eta_p^2 = .002$	H1a, not supported
Spatial embedding	1.98 (.15)	2.00 (.06)	$F(1,22,200) = 56.96, p < .001, \eta_p^2 = .003$	H1b, supported
Landscape of affective consciousness	.001 (.01)	.001 (.01)	$F(1,22,200) = .04, p = .85$	H1c, not supported
Landscape of cognitive consciousness	.001 (.02)	.001 (.02)	$F(1,22,200) = 1.56, p = .21$	H1d, not supported
Sentiment volatility	.21 (.08)	.20 (.08)	$F(1,22,200) = 4.59, p = .032, \eta_p^2 = .001$	H1e, supported
Progressive	235 (1.1%)	770 (3.5%)	$b = -.01, p = .92, \text{Wald} = .009$	RQ1a
Regressive	116 (.5%)	469 (2.1%)	$b = .21, p = .05, \text{Wald} = 3.94$	RQ1b
Comedy (U shaped)	153 (.7%)	523 (2.4%)	$b = .03, p = .72, \text{Wald} = .13$	RQ1c
Tragedy (Inverted U)	147 (.7%)	425 (1.9%)	$b = -.15, p = .13, \text{Wald} = 2.86$	RQ1d
Stable	4495 (20.2%)	14,666 (66.0%)	$b = .04, p = .35, \text{Wald} = .86$	RQ1e
Curiosity order drama	.59 (.28)	.59 (.28)	$F(1,22,200) = 1.47, p = .23$	RQ1f
Analytic	67.01 (14.35)	69.60 (15.41)	$F(1,22,200) = 121.71, p < .001, \eta_p^2 = .01$	
Clout	72.24 (9.76)	69.82 (10.48)	$F(1,22,200) = 216.33, p < .001, \eta_p^2 = .01$	
Tone	45.88 (13.19)	47.65 (13.70)	$F(1,22,200) = 68.22, p < .001, \eta_p^2 = .003$	

5.2.2 Predicting Shares Based on Narrativity. We conducted a negative binomial regression with the number of times the news article was shared as the dependent variable in our model and the focal narrativity elements as predictors, as in study 1. This model including the narrativity elements demonstrates a significant improvement in fit over the controls ($\Delta\chi^2(10) = 450.8, p < .001$; $\Delta\text{AIC} = 34.72$). The likelihood ratio test for the negative binomial regression model was significant ($\chi^2(17) = 21,51.65, p < .001$) indicating good fit. See table 6 for results.

A tragedy shape (inverted U) increased shares, supporting hypothesis 2j and replicating the findings of study 1. No other hypotheses were supported. Replicating the findings from study 1, a higher degree of landscape of affective consciousness and sentiment volatility decreased shares, as did a more negative tone. We discuss these findings and how they overlap with the findings of prior studies in the general discussion.

Table 6. Predicting Shares Based on Narrative Features, Study 3

Parameter	Model 1: Controls			Model 2: Main effects			Hypothesis Test
	B	SE	<i>p</i>	B	SE	<i>p</i>	
(Intercept)	5.55	.06	.00	5.68	.18	.00	
Word count	.01	.01	.00	.01	.01	.00	
Number of sentences	.01	.01	.00	.01	.01	.00	
Reliability (1 = real, 0 = fake)	-.51	.02	.00	-.51	.02	.00	
Analytic	-.03	.01	.00	-.03	.01	.00	
Clout	.01	.01	.00	-.01	.01	.00	
Tone	.01	.01	.00	-.01	.01	.00	
Article date	.01	.01	.00	-.01	.01	.00	
Temporal embedding				-.03	.03	.41	Hypothesis 2a not supported
Spatial embedding				.00	.08	.97	Hypothesis 2b not supported
Landscape of affective consciousness				-5.71	2.93	.05	Hypothesis 2c not supported
Landscape cognitive consciousness				-13.50	4.05	.00	Hypothesis 2d not supported
Sentiment volatility				-.29	.09	.00	Hypothesis 2e not supported
Curiosity drama order				.02	.02	.30	Hypothesis 2f not supported
Progressive				-.03	.03	.35	Hypothesis 2g not supported
Regressive				-.06	.04	.15	Hypothesis 2h not supported
Comedy (U shaped)				-.07	.04	.13	Hypothesis 2i not supported
Tragedy shape (Inverted U)				.09	.04	.03	Hypothesis 2j supported
Pearson X^2	63,066.80 (22,197)			62,616.00 (22187); X^2 Change (10) = 450.8, $p < .001$			
AIC	257,476.69			257,441.97; AIC change, 34.72			

6. General Discussion

The current work examines the prevalence of narrativity elements identified in prior consumer research (Berger et al., 2021; van Laer et al., 2019) in the context of real and fake news, using the computational approaches identified by previous researchers (H1 and RQ1). Across three studies, we also examine the relationship between narrative elements and sharing behavior or intentions (H2). In this section, we first compare findings across studies to identify reliable patterns (see also table 1), and then discuss our findings in terms of their theoretical and practical implications. We first discuss consistent findings across the studies and then consider the findings that were not replicated across studies.

6.1 Prevalence of Narrativity in Real and Fake News

Overall, the current work contributes broadly to the literature on fake news and word-of-mouth and to the literature on narrative persuasion (Escalas, 2007; Lien & Chen, 2013; Moradi & Badrinarayanan, 2021), specifically that which leverages automated text analysis (Berger et al., 2021; Moore & McFerran, 2017; van Laer et al., 2019). To our knowledge, we are the first to examine whether real and fake news differ in their narrativity. The present work compares real and fake news in terms of their spatial embedding, temporal embedding, landscapes of affective and cognitive consciousness, sentiment volatility, and story shapes (Hypothesis 1 and RQ1).

Study 1 found support for hypotheses 1a-1e: higher spatial embedding, temporal embedding, and landscapes of affective and cognitive consciousness in real versus fake news and higher sentiment volatility in fake versus real news. Two of these relationships were robust to context and replicated in study 3: sentiment volatility was higher in fake news, and spatial embedding was more prevalent in real news. This finding suggests that these elements of narrativity may be the most fruitful in identifying real versus fake news. As previously described, fake news writers' motives center on attracting and sustaining attention (Plangger & Campbell,

2022; Zhang & Ghorbani, 2020), which may be achieved by incorporating more period-to-period shifts in emotional variation (or sentiment volatility). Indeed, past work suggests that content variety reduces boredom (Ratner, Kahn, & Kahneman, 1999) and can make an experience feel more exhilarating (Etkin & Mogilner, 2016) and possibly reduce the audience's ability to consider whether the content is fake, which is aligned with the goals of fake news writers. Relatedly, fake news writers may be less inclined than real news writers to use explaining language designed to foster understanding, which is consistent with the finding that real news features a higher landscape of cognitive consciousness. Additionally, while not a focal part of the paper, fake news in both studies was more negative in tone, replicating a well-established effect (Damstra et al., 2021; Gupta et al., 2022), and was higher in clout (a more external focus, exemplified by using more second-person singular pronouns). This finding is aligned with our suggestion that fake news writers are less likely to use explaining language about the causes of events and more likely to focus on external manifestations that attract attention. Further, fake news writers' efforts to appear "objective" and fact-based may result in more externally oriented language.

The other hypothesized elements of narrativity either did not differ between real and fake news in study 3 (landscape of affective and cognitive consciousness) or were significant in the opposite direction (temporal embedding). In terms of RQ1, while study 1 found that progressive and comedy (U shapes) genres were more prevalent in fake news articles and stable shapes were more prevalent in real news articles, there were no differences in article shapes in study 3.

There are two possible interpretations of the results that failed to replicate across studies: these characteristics are not reliable in distinguishing real versus fake news, or they are only reliable in certain contexts. For instance, when the news focuses more prevalently on people

(celebrities versus health or politics), the explaining language associated with a person's thinking and feeling (landscapes of affective and cognitive consciousness) may be equally present regardless of whether the news is real or fake, or may reflect writers' beliefs about the audience's (relatively higher) level of familiarity with the characters featured in the stories. Future work remains to address the question of domain specificity.

6.2 Association Between Narrativity and Sharing

The current work contributes to the word-of-mouth literature by examining how narrativity influences the outcome of content sharing, an outcome of increasing importance to marketers and businesses (Babić Rosario et al., 2016; 2020). This work also contributes to the literature on narrative persuasion (Escalas, 2007; Lien & Chen, 2013; Moradi & Badrinarayanan, 2021). While prior work has examined how sentiment volatility (Berger et al., 2021) and other elements of narrativity (van Laer et al., 2019) are associated with attitude toward the content, media sharing is a distinct outcome variable. The present work examines the effects of spatial embedding, temporal embedding, landscapes of affective and cognitive consciousness, sentiment volatility, and story shapes on news sharing (hypothesis 2).

Across three datasets examining the outcome of actual shares and sharing intentions, the only hypothesis that reliably replicated is hypothesis 2j: tragedy shapes are associated with more sharing behavior. Out of all the shapes, tragedy shapes perhaps best leverage the motives people have for sharing content online. Prior research shows that observers judge someone who shares negative (vs. positive) information as more intelligent (Amabile, 1983). However, people also avoid sharing negative content to avoid seeming like a negative person (Berger, 2014; Berger & Milkman, 2012). Tragedy shapes, containing an initially positive and then negative trajectory, may effectively balance these competing desires. Similarly, tragedy shapes may tap into

consumers' desires to help others. By sharing news in tragedy shapes, people may be motivated to help others avoid a similar fate (Lovett, Peres, & Shachar, 2013). Finally, tragedy shapes may simply be the most evocative. As mentioned above, social media content may distract people from prioritizing truth when they decide what to share (Chen et al., 2021), favoring aspects that they perceive as more likely to engage others – which tragedy shapes may exemplify. Emotion regulation (specifically, managing negative emotions) has been identified as one reason why people share content online (Berger, 2014), and news articles ending in a negative manner may be particularly likely to arouse this need.

However, there were other consistent effects, but in the opposite of the proposed direction: sentiment volatility *decreased* sharing intentions or shares. These findings are particularly noteworthy, as there is much more research examining what causes people to share than what causes them *not* to share, both in the word-of-mouth (Berger, 2014; Berger & Milkman, 2012; Babić Rosario et al., 2020) and fake news literature (Pennycook & Rand, 2021).

Sentiment volatility is similar to tragedy shapes in that these constructs focus on or feature sentiment variation, but sentiment volatility captures the *amount* of variation. Our finding is consistent with past research that shows that consumers' experience of too much vacillation can be uncomfortable (van Harreveld, Van der Pligt, & De Liver, 2009), reducing persuasion. Prior research also shows that the experience of novelty or arousal past a set point can have diminishing effects on persuasion-related outcomes (Grant & Schwartz, 2011).

The landscape of affective consciousness also consistently decreased sharing intentions and shares. This construct represents the extent to which a text recounts initial events about which characters express feelings that lead to subsequent events, and is, in effect, emotionally based explaining language. While a reader's experience of emotional arousal may influence

sharing behavior through unconscious or affective mechanisms (Berger, 2014; Rimé, 2009) – which may be evoked based on text content or their idiosyncratic response to text content – readers may interpret an author’s use of emotion in describing a character’s actions through a more cognitive, conscious process. Specifically, people may view news articles that write about a person’s emotional (versus cognitive) response to an event, which guides their subsequent actions, as less newsworthy or less credible. For instance, people are more likely to trust others that exhibit self-control or are less likely to act on their emotions (Righetti & Finkenauer, 2011). Alternatively, audiences may experience language with greater landscape of affective consciousness as more private or intimate, which may reduce their propensity to share.

All other relationships were either not significant or significant in the opposite direction (e.g., landscape of cognitive consciousness increased shares in studies 1 and 2 but decreased them in study 3). As mentioned above, there are two possible interpretations for the results that failed to replicate: these characteristics do not reliably influence sharing behavior, or they are only reliable in certain contexts, which future work remains to address. Finally, it is possible that fake news writers used natural language processing tools driven by artificial intelligence, intentionally mimicking the features of a real news article and reducing their distinguishability on certain characteristics.

6.3 Applied Implications

The potential impact of fake news on consumers and firms is substantial. Directly, fake news can negatively influence consumers’ beliefs about brands and products when mentioned in news articles (Flostrand et al., 2019) and can adversely impact business sales and stock prices (Atkinson, 2019). Indirectly, consumers develop negative attitudes toward advertisements for products that appear alongside fake news (Visentin et al., 2019). More broadly, the prevalence of

fake news has important implications for consumer trust in the media and technology-based platforms in which consumers and businesses are embedded (Gupta et al., 2022).

From an applied standpoint, this work contributes to the efforts of media gatekeepers, governments, and consumer advocates to identify and curb the effects of fake news. There are many initiatives that are currently underway to develop algorithms that can help identify potential fake news on social media platforms such as Facebook (Albahr & Albahr, 2020), and the current work provides concrete direction about aspects of linguistic content that can be measured and incorporated into such identification algorithms. Specifically, the current work showed that spatial embedding was more prevalent in real news, and sentiment volatility was higher in fake news.

The current work also addresses the question of what aspects of news content contribute to the spread of news, which is relevant to marketers more broadly. Three features of narrative content reliably influenced media sharing behavior and intentions: a tragedy shape increased shares, while sentiment volatility and the landscape of affective consciousness decreased shares. The present findings are thus relevant for content creators who disseminate their news on online platforms and whose goal is to increase sharing. Such creators might consider writing negative news in a tragedy (rather than purely negative) format. Similarly, authors should avoid constructing articles that feature too much period-to-period variation, despite a potential temptation to do so in an effort to attract attention. While prior work has shown that emotionality can enhance content sharing (Berger, 2014), the present work suggests that content creators seeking to encourage shares should avoid explaining how a person's feelings caused subsequent events (landscape of affective consciousness).

Finally, while upstream efforts to first identify and flag fake news are central to addressing any social problem, downstream efforts to inform and empower consumers to protect themselves are also important. Accordingly, the current work could inform current efforts to educate the public about the characteristics of fake news (Maertens et al., 2021), with the goal of aiding consumers in identifying (and refusing to disseminate) fake news.

6.4 Limitations and Future Research

One limitation of the present work relates to the datasets used and in the ways “real” and “fake” news were distinguished. In two of the datasets, truth labels were identified by cross referencing the fact-checking site PolitiFact, while in two of the other datasets, truth labels were determined at the publisher level. Future work can examine whether the effects found in the current work generalize across fake news sources as labeled by alternative methods. In addition, the datasets included in the current work focus on political, health, and celebrity-related news. As mentioned above, several elements of narrativity were not significant in distinguishing between real and fake news or predicting shares. It is possible that these characteristics are truly not reliable in distinguishing real versus fake news and predicting media sharing behavior, or they are only reliable in certain contexts. Furthermore, the effects we established as reliable in the examined contexts may not generalize to others.

Relatedly, one of the limitations of studies 1 and 3 is associated with the outcome of “shares” as captured in the secondary datasets from Twitter: Twitter has undergone a number of changes over the years that could impact participants’ likelihood of sharing the news article. While we controlled for publication date in studies 1 and 3 and measured participants’ sharing intentions directly in study 2, future work remains to replicate these findings with alternative measures of sharing (e.g., emailing the media content to others). An additional limitation is that

we measured sharing intentions (rather than actual sharing behavior) in study 2. While behavioral intentions are often correlated with behavior, there are obviously situations in which stated intentions diverge from actual behavior.

Future work can examine whether people have lay associations between the presence of narrativity in a news article and their judgments of whether the article is real or fake. For instance, consumers may believe that a high degree of sentiment volatility signals fake news, and a high degree of cognitive consciousness signals real news. If this is the case, then fake news mimicking real news may be better able to blend in as real news, while consumers may incorrectly reject real news containing certain narrative features as fake.

Future work may also adopt an experimental approach to examine some of the findings of the current work in more detail. For instance, we found that fake news was higher in sentiment volatility, but sentiment volatility was associated with lower levels of social sharing. One might question why fake news authors might employ a content strategy that is ineffective. A potential answer is that these authors are not aware that their efforts to incorporate moment-to-moment variability are hampering their efforts to spread their content. Alternatively, sentiment volatility may enhance sharing in contexts or among consumers not examined in the current work. Future research can examine whether the specific content of the message or consumer's goal in consuming the content are moderators of the relationship between sentiment volatility and sharing, such as more entertainment-focused media (versus news content) or a goal to be entertained (versus a goal to be informed). Furthermore, this research could reconcile differences between the current work's finding that sentiment volatility decreases shares but Berger et al.'s (2021) finding that it enhances liking; the context and consumers' consumption motive may drive these differences.

Overall, the current work showed that real and fake news vary in terms of their narrativity and that differences in the presence of these characteristics predict the likelihood that the article is shared on social media. As fake news is one of the most pressing concerns for our current society, we hope the current work opens the door to subsequent research on the topic.

6.6 Conclusion

The present work compares real and fake news in terms of their spatial embedding, temporal embedding, landscapes of affective and cognitive consciousness, sentiment volatility, and story shapes. Across studies, there was more sentiment volatility in fake news, and spatial embedding was more prevalent in real news. The present work also examined how elements of narrativity predict the audience's sharing behavior or intentions. Tragedy shapes (in which events improve and then decline) are associated with more news sharing behavior, whereas higher amounts of sentiment volatility and landscape of affective consciousness decreases sharing intentions or shares.

Citations

- Ahmadi, I., Waltenrath, A., & Janze, C. (2023). Congruency and Users' Sharing on Social Media Platforms: A Novel Approach for Analyzing Content. *Journal of Advertising*, 52(3), 369-386.
- Albahr, A., & Albahr, M. (2020). An empirical comparison of fake news detection using different machine learning algorithms. *International Journal of Advanced Computer Science and Applications*, 11(9), 146-152.

- Aleti, T., Pallant, J. I., Tuan, A., & Van Laer, T. (2019). Tweeting with the stars: Automated text analysis of the effect of celebrity social media communications on consumer word of mouth. *Journal of Interactive Marketing*, 48(1), 17-32.
- Amabile, T. M. (1983). Brilliant but cruel: Perceptions of negative evaluators. *Journal of Experimental Social Psychology*, 19(2), 146-156.
- Arvidsson, A., & Caliendo, A. (2016). Brand public. *Journal of Consumer Research*, 42(5), 727-748.
- Asubiaro, T. V., & Rubin, V. L. (2018). Comparing features of fabricated and legitimate political news in digital environments (2016-2017). *Proceedings of the Association for Information Science and Technology*, 55(1), 747-750.
- Atkinson, C. (2019). Fake news can cause irreversible damage to companies—and sink their stock prices. Business News. NBC news.
- Babić Rosario, A., Sotgiu, F., De Valck, K., & Bijmolt, T. H. (2016). The effect of electronic word of mouth on sales: A meta-analytic review of platform, product, and metric factors. *Journal of Marketing Research*, 53(3), 297-318.
- Babić Rosario, A., De Valck, K., & Sotgiu, F. (2020). Conceptualizing the electronic word-of-mouth process: What we know and need to know about eWOM creation, exposure, and evaluation. *Journal of the Academy of Marketing Science*, 48, 422-448.
- Berger, J. (2014). Word of mouth and interpersonal communication: A review and directions for future research. *Journal of Consumer Psychology*, 24(4), 586-607.
- Berger, J., & Milkman, K. L. (2012). What makes online content viral? *Journal of Marketing Research*, 49(2), 192-205.

- Berger, J., Kim, Y. D., & Meyer, R. (2021). What makes content engaging? How emotional dynamics shape success. *Journal of Consumer Research*, 48(2), 235-250.
- Bergkvist, L., & Rossiter, J. R. (2007). The predictive validity of multiple-item versus single-item measures of the same constructs. *Journal of Marketing Research*, 44(2), 175-184.
- Berthon, P. R., & Pitt, L. F. (2018). Brands, truthiness and post-fact: managing brands in a post-rational world. *Journal of Macromarketing*, 38(2), 218-227.
- Born, K., & Edgington, N. (2017). Analysis of philanthropic opportunities to mitigate the disinformation/propaganda problem. Retrieved from <https://www.hewlett.org/wp-content/uploads/2017/11/Hewlett-Disinformation-Propaganda-Report.pdf>
- Bradshaw, S., Howard, P. N., Kollanyi, B., & Neudert, L. M. (2020). Sourcing and automation of political news and information over social media in the United States, 2016-2018. *Political Communication*, 37(2), 173-193.
- Bruner, J. S. (2009). *Actual minds, possible worlds*. Harvard University Press.
- Buccoliero, L., Bellio, E., Crestini, G., & Arkoudas, A. (2020). Twitter and politics: Evidence from the US presidential elections 2016. *Journal of Marketing Communications*, 26(1), 88-114.
- Buchanan, T. (2020). Why do people spread false information online? The effects of message and viewer characteristics on self-reported likelihood of sharing social media disinformation. *Plos One*, 15(10), e0239666.
- Casero-Ripollés, A., Marcos-García, S., & Alonso-Muñoz, L. (2020). New formats for local journalism in the era of social media and big data: From transmedia to storytelling. *Journalistic Metamorphosis: Media Transformation in the Digital Age*, 70, 69-83.

- Centeno, D., & Wang, J. J. (2017). Celebrities as human brands: An inquiry on stakeholder-actor co-creation of brand identities. *Journal of Business Research*, 74(May), 133-138.
- Chen, X., Pennycook, G., & Rand, D. (2023). What makes news sharable on social media? *Journal of Quantitative Description: Digital Media*, 3.
- Damstra, A., Boomgaarden, H. G., Broda, E., Lindgren, E., Strömbäck, J., Tsfat, Y., & Vliegenthart, R. (2021). What does fake look like? A review of the literature on intentional deception in the news and on social media. *Journalism Studies*, 22(14), 1947-1963.
- Di Domenico, G., Sit, J., Ishizaka, A., & Nunan, D. (2021). Fake news, social media and marketing: A systematic review. *Journal of Business Research*, 124, 329-341.
- Dudukovic, N. M., Marsh, E. J., & Tversky, B. (2004). Telling a story or telling it straight: The effects of entertaining versus accurate retellings on memory. *Applied Cognitive Psychology*, 18(2), 125-143.
- Escalas, J. E. (2004). Imagine yourself in the product: Mental simulation, narrative transportation, and persuasion. *Journal of Advertising*, 33(2), 37-48.
- Escalas, J. E. (2007). Self-referencing and persuasion: Narrative transportation versus analytical elaboration. *Journal of Consumer Research*, 33(4), 421-429.
- Etkin, J., & Mogilner, C. (2016). Does variety among activities increase happiness? *Journal of Consumer Research*, 43(2), 210-229.
- Fivush, R., Thorstad, R., & Graci, M. E. (2021). Similarity of personal and cultural narratives. Unpublished manuscript, Emory University, GA.

- Flostrand, A., Pitt, L., & Kietzmann, J. (2020). Fake news and brand management: a Delphi study of impact, vulnerability and mitigation. *Journal of Product & Brand Management*, 29(2), 246-254.
- Fludernik, M. (2009). *An Introduction to Narratology*. Routledge.
- Gergen, K. J., & Gergen, M. M. (1988). Narrative and the self as relationship. In *Advances in Experimental Social Psychology* (Vol. 21, pp. 17-56). Academic Press.
- Grant, A. M., & Schwartz, B. (2011). Too much of a good thing: The challenge and opportunity of the inverted U. *Perspectives on Psychological Science*, 6(1), 61-76.
- Graves, L., Nyhan, B., & Reifler, J. (2016). Understanding innovations in journalistic practice: A field experiment examining motivations for fact-checking. *Journal of Communication*, 66(1), 102-138.
- Green, M. C., & Brock, T. C. (2000). The role of transportation in the persuasiveness of public narratives. *Journal of Personality and Social Psychology*, 79(5), 701-721.
- Gupta, A., Li, H., Farnoush, A., & Jiang, W. (2022). Understanding patterns of COVID infodemic: A systematic and pragmatic approach to curb fake news. *Journal of Business Research*, 140 (February), 670-683.
- Hennig-Thurau, T., Wiertz, C., & Feldhaus, F. (2015). Does Twitter matter? The impact of microblogging word of mouth on consumers' adoption of new movies. *Journal of the Academy of Marketing Science*, 43, 375-394.
- Hwong, Y. L., Oliver, C., Van Kranendonk, M., Sammut, C., & Seroussi, Y. (2017). What makes you tick? The psychology of social media engagement in space science communication. *Computers in Human Behavior*, 68, 480-492.
- Ireton, C., & Posetti, J. (2018). *Journalism, Fake News & Disinformation: Handbook for*

- Journalism Education and Training*. Unesco Publishing.
- Jalali, N. Y., & Papatla, P. (2019). Composing tweets to increase retweets. *International Journal of Research in Marketing*, 36(4), 647-668.
- Kotler, P., & Levy, S. J. (1969). Broadening the concept of marketing. *Journal of Marketing*, 33(1), 10-15.
- Koohikamali, M., & Sidorova, A. (2017). Information Re-Sharing on Social Network Sites in the Age of Fake News. *Informing Science*, 20, 215-235.
- Lien, N. H., & Chen, Y. L. (2013). Narrative ads: The effect of argument strength and story format. *Journal of Business Research*, 66(4), 516-522.
- Lovett, M. J., Peres, R., & Shachar, R. (2013). On brands and word of mouth. *Journal of Marketing Research*, 50(4), 427-444.
- Marchal, N., Kollanyi, B., Neudert, L. M., & Howard, P. N. (2019). Junk news during the EU parliamentary elections: Lessons from a seven-language study of Twitter and Facebook. *University of Oxford*.
- Maertens, R., Roozenbeek, J., Basol, M., & van der Linden, S. (2021). Long-term effectiveness of inoculation against misinformation: Three longitudinal experiments. *Journal of Experimental Psychology: Applied*, 27(1), 1.
- McDermott, R. (2019). Psychological underpinnings of post-truth in political beliefs. *PS: Political Science & Politics*, 52(2), 218-222.
- Moore, S. G., & McFerran, B. (2017). She said, she said: Differential interpersonal similarities predict unique linguistic mimicry in online word of mouth. *Journal of the Association for Consumer Research*, 2(2), 229-245.

- Moradi, M., & Badrinarayanan, V. (2021). The effects of brand prominence and narrative features on crowdfunding success for entrepreneurial aftermarket enterprises. *Journal of Business Research*, 124, 286-298.
- Nahari, G. (2018). Reality monitoring in the forensic context: Digging deeper into the speech of liars. *Journal of Applied Research in Memory and Cognition*, 7(3), 432-440.
- National Science Foundation. (2022). Inviting proposals related to information integrity to the Secure and Trustworthy Cyberspace Program. Retrieved from https://beta.nsf.gov/funding/opportunities/inviting-proposals-related-information-integrity-secure-and-trustworthy_.
- Oh, S., & Syn, S. Y. (2015). Motivations for sharing information and social support in social media: A comparative analysis of Facebook, Twitter, Delicious, YouTube, and Flickr. *Journal of the Association for Information Science and Technology*, 66(10), 2045-2060.
- Oschatz, C., & Marker, C. (2020). Long-term persuasive effects in narrative communication research: A meta-analysis. *Journal of Communication*, 70(4), 473-496.
- Paharia, N., Keinan, A., Avery, J., & Schor, J. B. (2011). The underdog effect: The marketing of disadvantage and determination through brand biography. *Journal of Consumer Research*, 37(5), 775-790.
- PBS. (2021, September 29). When sensationalism became fake news. Retrieved from <https://www.pbs.org/wgbh/americanexperience/features/conversations-when-sensationalism-became-fake-news/>
- Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). *The development and psychometric properties of LIWC2015*.

- Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. *Trends in Cognitive Sciences*, 25(5), 388-402.
- Pezzuti Roberts, T. (2021). Certainty in Language Increases Consumer Engagement on Social Media. *Journal of Interactive Marketing*, 53(1), 32-46.
- Plangger, K., & Campbell, C. (2022). Managing in an era of falsity: Falsity from the metaverse to fake news to fake endorsement to synthetic influence to false agendas. *Business Horizons*, 65, 713-717.
- Quesenberry, K. A., & Coolson, M. K. (2019). Drama goes viral: Effects of story development on shares and views of online advertising videos. *Journal of Interactive Marketing*, 48, 1-16.
- Ratner, R. K., Kahn, B. E., & Kahneman, D. (1999). Choosing less-preferred experiences for the sake of variety. *Journal of Consumer Research*, 26(1), 1-15.
- Reagan, A. J., Mitchell, L., Kiley, D., Danforth, C. M., & Dodds, P. S. (2016). The emotional arcs of stories are dominated by six basic shapes. *EPJ Data Science*, 5(1), 1-12.
- Reinemann, C., Stanyer, J., Scherr, S., & Legnante, G. (2012). Hard and soft news: A review of concepts, operationalizations and key findings. *Journalism*, 13(2), 221-239.
- Righetti, F., & Finkenauer, C. (2011). If you are able to control yourself, I will trust you: the role of perceived self-control in interpersonal trust. *Journal of Personality and Social Psychology*, 100(5), 874-886.
- Rimé, B. (2009). Emotion elicits the social sharing of emotion: Theory and empirical review. *Emotion review*, 1(1), 60-85.
- Roeh, I. (1989). Journalism as storytelling, coverage as narrative. *American Behavioral Scientist*, 33(2), 162-168.

- Illari, L., Restrepo, N. J., & Johnson, N. F. (2023). Rise of post-pandemic resilience across the distrust ecosystem. *arXiv preprint arXiv:2307.11393*.
- Ruffo, G., Semeraro, A., Giachanou, A., & Rosso, P. (2023). Studying fake news spreading, polarisation dynamics, and manipulation by bots: A tale of networks and language. *Computer Science Review*, 47, 100531.
- Schwarz, N., & Jalbert, M. (2020). When (fake) news feels true: Intuitions of truth and the acceptance and correction of misinformation. In *The Psychology of Fake News* (pp. 73-89). Routledge.
- Shrestha, A., Duran, J., Spezzano, F., & Serra, E. (2023). Joint Credibility Estimation of News, User, and Publisher via Role-Relational Graph Convolutional Networks. *ACM Transactions on the Web*.
- Shu, K. (2019). FakeNewsNet. Harvard Dataverse, V2. doi:10.7910/DVN/UEMMHS
- Shu, K., Mahudeswaran, D., Wang, S., Lee, D., & Liu, H. (2020). Fakenewsnet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media. *Big Data*, 8(3), 171-188.
- Singh, V. K., Ghosh, I., & Sonagara, D. (2021). Detecting fake news stories via multimodal analysis. *Journal of the Association for Information Science and Technology*, 72(1), 3-17.
- Sporer, S. L. (1997). The less travelled road to truth: Verbal cues in deception detection in accounts of fabricated and self-experienced events. *Applied Cognitive Psychology: The Official Journal of the Society for Applied Research in Memory and Cognition*, 11(5), 373-397.

- Tamul, D. J., Ivory, A. H., Hotter, J., & Wolf, J. (2020). All the President's tweets: effects of exposure to Trump's "Fake News" accusations on perceptions of journalists, news stories, and issue evaluation. *Mass Communication and Society*, 23(3), 301-330.
- Thorsen, E., & Jackson, D. (2018). Seven characteristics defining online news formats: Towards a typology of online news and live blogs. *Digital Journalism*, 6(7), 847-868.
- Van Harreveld, F., Van der Pligt, J., & De Liver, Y. N. (2009). The agony of ambivalence and ways to resolve it: Introducing the MAID model. *Personality and Social Psychology Review*, 13(1), 45-61.
- Van Krieken, K., & Sanders, J. (2021). What is narrative journalism? A systematic review and an empirical agenda. *Journalism*, 22(6), 1393-1412.
- Van Laer, T., De Ruyter, K., Visconti, L. M., & Wetzels, M. (2014). The extended transportation-imagery model: A meta-analysis of the antecedents and consequences of consumers' narrative transportation. *Journal of Consumer Research*, 40(5), 797-817.
- Van Laer, T., Edson Escalas, J., Ludwig, S., & Van Den Hende, E. A. (2019). What happens in Vegas stays on TripAdvisor? A theory and technique to understand narrativity in consumer reviews. *Journal of Consumer Research*, 46(2), 267-285.
- Visentin, M., Pizzi, G., & Pichierri, M. (2019). Fake news, real problems for brands: The impact of content truthfulness and source credibility on consumers' behavioral intentions toward the advertised brands. *Journal of Interactive Marketing*, 45(1), 99-112.
- Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146-1151.

- Wardle, C. (2006). "IT COULD HAPPEN TO YOU" The move towards "personal" and "societal" narratives in newspaper coverage of child murder, 1930–2000. *Journalism Studies*, 7(4), 515-533.
- Yoon, G., Li, C., Ji, Y., North, M., Hong, C., & Liu, J. (2018). Attracting comments: Digital engagement metrics on Facebook and financial performance. *Journal of Advertising*, 47(1), 24-37.
- Zaman, M., Vo-Thanh, T., Nguyen, C. T., Hasan, R., Akter, S., Mariani, M., & Hikkerova, L. (2023). Motives for posting fake reviews: Evidence from a cross-cultural comparison. *Journal of Business Research*, 154, 113359.
- Zhang, X., & Ghorbani, A. A. (2020). An overview of online fake news: Characterization, detection, and discussion. *Information Processing & Management*, 57(2), 102025.
- Zhou, X., Jain, A., Phoha, V. V., & Zafarani, R. (2020). Fake news early detection: A theory-driven model. *Digital Threats: Research and Practice*, 1(2), 1-25.
- Zhou, X., Mulay, A., Ferrara, E., & Zafarani, R. (2020, October). Recovery: A multimodal repository for covid-19 news credibility research. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management* (pp. 3205-3212).
- Zhang, X., Cao, J., Li, X., Sheng, Q., Zhong, L., & Shu, K. (2021, April). Mining dual emotion for fake news detection. In *Proceedings of the Web Conference 2021* (pp. 3465-3476).